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EIGHTH REVIEW UNDER THE EXTENDED CREDIT FACILITY ARRANGEMENT, REQUESTS FOR A WAIVER OF NONOBSERVANCE OF PERFORMANCE CRITERION, AND FINANCING ASSURANCES REVIEW—DEBT SUSTAINABILITY ANALYSIS

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Risk of external debt distress	High
Overall risk of debt distress	High
Granularity in the risk rating	Sustainable
Application of judgment	No

Sierra Leone's public debt is sustainable but risks of external and overall debt distress are high, unchanged from the previous DSA published in June 2023.[1] *Last year's large fiscal overruns and the sharp depreciation of the leone have resulted in a deterioration of both solvency and liquidity indicators. The external debt service-to-revenue ratio and the PV of the public debt-to-GDP ratio exhibit larger breaches of their respective thresholds. In addition, the overall debt-service to revenue ratio and the PV of external debt-to-GDP indicator have shifted up notably. A larger and more frontloaded fiscal adjustment is now urgently needed to ensure that debt remains sustainable. All debt indicators are on a declining trend over the medium to long-term. However, this is predicated on steadfast implementation of the planned fiscal adjustment, continued reliance on concessional financing and grants, and the projected growth recovery. Efforts to seek further grant financing and lengthen the maturity structure of debt remain key as debt service ratios and gross financing needs will remain elevated over the medium and long-term. Domestic rollover risks are attenuated by high interest margins, limited alternative investment options for domestic banks and the authorities' commitment to limit future domestic borrowing. Sierra Leone is susceptible to growth and exchange rate shocks as well as fiscal slippages, all of which present downside risks to the debt outlook.*

[1] Sierra Leone's debt-carrying capacity remains rated "medium" with a composite indicator value of 2.73 based on the April 2023 IMF's World Economic Outlook (WEO) and the 2022 World Bank's Country Policy and Institutional Assessment (CPIA).

PUBLIC DEBT COVERAGE

1. Public debt coverage remains the same as in the previous DSA. The debt stock includes central government public and publicly guaranteed debts on a residency basis (Text Table 1). The DSA also includes the latest estimate of the consolidated stock of domestic payment arrears. The government is working, with the support of development partners including the World Bank through the Sustainable Development Finance Policy (SDFP), to improve its financial and debt management system, and to enhance the accounting and timely reporting of public debt, including the debts of state-owned enterprises (SOEs) and self-accounting-bodies. This DSA uses the debt stock data as of end-2022.

Text Table 1. Public Sector Debt Coverage Under the Baseline Scenario

	Subsectors of the public sector	Sub-sectors covered
1	Central government	X
2	State and local government	
3	Other elements in the general government	
4	o/w: Social security fund	X
5	o/w: Extra budgetary funds (EBFs)	
6	Guarantees (to other entities in the public and private sector, including to SOEs)	X
7	Central bank (borrowed on behalf of the government)	X
8	Non-guaranteed SOE debt	

Note: Blank indicates there is no data.

Text Table 2. Coverage of the Contingent Liabilities' Stress Test

1. The country's coverage of public debt	The central government plus social security, central bank, government-guaranteed debt		
	Default	**Used for the Analysis**	**Reasons for deviations from the default setting**
2. Other elements of the general government not captured in 1.	0 percent of GDP	0.0	Reflect the authorities' estimate of total indebtedness of SOEs.
3. SOE's debt (guaranteed and not guaranteed by the government) 1/	2 percent of GDP	7.0	
4. PPP	35 percent of PPP stock	0.0	
5. Financial market (the default value of 5 percent of GDP is the minimum value)	5 percent of GDP	5.0	
Total (2+3+4+5) (in percent of GDP)		12.0	

1/ The default shock of 2 percent of GDP will be triggered for countries whose government guaranteed debt is not fully captured under the country's public debt definition (1). If it is already included in the government debt (1) and risks associated with SOE's debt not guaranteed by the government is assessed to be negligible, a country team may reduce this to 0 percent.

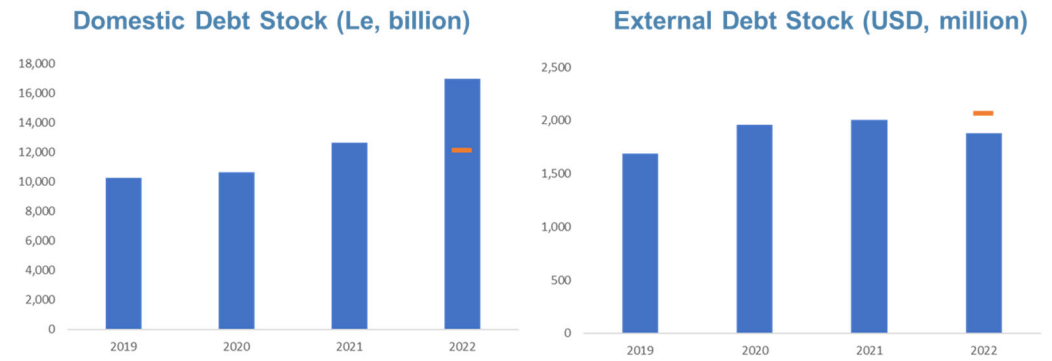
2. The assumptions on the contingent liability stress test remain the same as in the previous DSA. Total contingent liabilities are estimated at 12 percent of GDP (Text Table 2). Specifically, (i) the contingent liability for SOE debt is set at 7 percent of GDP, higher than the default value reflecting the authorities' latest estimates; (ii) contingent liabilities from financial markets are set at the standard minimum value of 5 percent of GDP based on the average fiscal cost

of a financial crisis in LICs since 1980; and (iii) the contingent liability of other elements of the general government is set at 0 percent of GDP because the baseline reflects estimated domestic arrears.

BACKGROUND ON DEBT

3. End-2022 stock data shows that Sierra Leone's public debt continued its increasing trend. While external debt turned out to be lower than projected in US dollar terms in 2022, total public debt rose to 93 percent of GDP in 2022 (from 79 percent of GDP in 2021) due to (i) the sharp depreciation of the leone (67.3 percent yoy depreciation at end-2022), and (ii) the high 2022 fiscal deficit, which led to substantial additional domestic borrowing, largely through 364-day T-bills (Text Figure 1). The authorities maintained new external borrowing below the Fund-supported program ceiling.

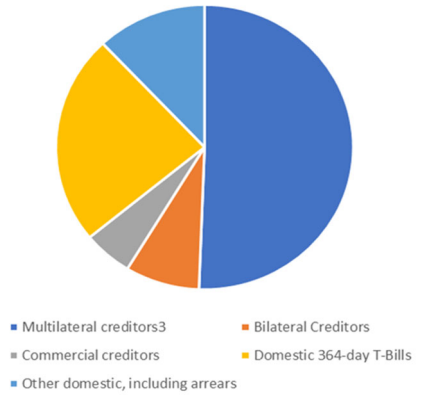
Text Figure 1. Debt Stock (Domestic and External)



Note: Orange markers in 2022 show the projection as of the June 2022 DSA.

4. The composition of public debt is largely unchanged from the previous DSA. Approximately 67 percent of total public and publicly guaranteed (PPG) debt is external debt, of which about 80 percent comprised non-restructurable obligations to multilateral creditors including the IMF (26 percent of total PPG debt stock) and the WB (24 percent of total PPG debt stock). On the domestic debt side, around 60 percent of domestic debt is owed to commercial banks mainly in the form of 364-day T-bills. The remainder is distributed between the BSL, the non-bank sector, and domestic arrears owed to suppliers (Text Figure 2).

Text Figure 2. Debt Stock as of End-2022



BACKGROUND ON MACRO FORECASTS

5. The macroeconomic assumptions underlying this DSA have remained broadly unchanged from the June 2023 DSA (Text Table 3).

Text Table 3. Macroeconomic Assumptions

	Current DSA			Previous DSA (June 2023)		
	2023	2033	LT ave.1/	2023	2033	LT ave.
Real GDP growth (in percent)	2.7	4.6	4.5	2.7	4.6	4.5
Inflation (GDP deflator, in percent)	37.1	6.4	12.1	34.7	6.5	12.6
Primary deficit (percent of GDP)	1.7	1.0	0.5	1.4	0.9	0.4
Non-interest current account deficit (percent of GDP)	5.6	4.9	3.9	6.7	5.5	4.0

1/ Average between 2023 to 2033.

- **Growth.** In 2023, growth is projected to decelerate to 2.7 percent (from 3.5 percent in 2022) amid the strong fiscal consolidation, and as rapidly rising prices are expected to weigh on domestic demand. The ramp-up of iron ore production at the Marampa and Tonkolili mines should boost production; supportive government policies to provide seed rice and implements to farmers should support agricultural output. Growth in 2024 is expected to accelerate due to: continued expansion in mining activity, and sustained improvements in agriculture. In addition, export shipments are predicted to grow (while iron ore prices moderate), and the macroeconomic stabilization is expected to bolster domestic demand. Growth is projected to reach Sierra Leone's long-run potential of about 4½ percent over the medium term, supported by buoyant activity in the mining and agricultural sectors. The tighter macroeconomic policies should help create room for private sector led growth.[2] Downside risks to the growth outlook are significant. Political tensions, food insecurity, and recent regulatory price increases could fuel social unrest, undermine investor confidence, and stymie the policy agenda. A failure to achieve the programmed adjustment could lead to difficulties in financing the deficit, reinvigorate deposit dollarization and intensify debt related risks. A global slowdown, tighter financial conditions, a more protracted Russia's war in Ukraine, and geographical fragmentation could weigh on external demand.

- **Inflation.** Inflation, as measured by the GDP deflator, is projected at 44.3 percent in 2023. The measure is evolving broadly in line with, but is currently lower than, the expected average consumer price inflation of 45.5 percent for 2023. The upward revision in inflation reflects recent adjustments of administrative prices for fuel and electricity tariffs, continued increases in food prices, and the high pass-through of the sharp Leone depreciation in the first half of the year on domestic prices. Base money growth is projected to contract by 17.1 percent in real terms in

[2] The ongoing rebasing of the national accounts will help improve the GDP estimates and computation of related deflators.

2023. The BSL has raised the policy rate by 250 bps since July 2023 in a clear signal of its intention to contain inflation. Over the medium-term, inflation is projected to gradually decline to single digits amid the tighter macroeconomic policy stance.

- **Fiscal.** In 2023, the overall fiscal deficit is projected to narrow to 5.8 percent of GDP from -10.5 percent in 2022, modestly higher than the 5.6 ppt of GDP deficit target for 2023 in the staff report for the 6th/7th reviews of IMF ECF arrangement due to revenue shortfalls.[3] Revenues are expected to be boosted by administrative measures and tax policy measures. Tax policy measures in the 2023 and 2024 Finance Acts alone are expected to raise annual revenues by around 3 ppts of GDP over time (the full yield will take time to materialize). Tax policy measures in the 2023 Finance Act include the adoption of a minimum alternative tax, the removal of exemptions on GST, and the harmonization of excise tax rates for local and imported excisable products; measures in the 2024 Finance Act include broadening the GST base, adopting a specific excise tax on plastics, restoring duties on rice imports, introducing an education levy, and harmonizing the final withholding tax rates for capital income (excluding capital gains tax). Spending is projected to fall by 5.9 ppts of GDP in 2023. Measures on spending include canceling further allocations to MDAs in Q4-2023 (0.2 ppts of GDP), which in addition to the removal of one-offs yields about 1.7 ppts of GDP relative to 2022; (ii) reducing remaining subsidies to EDSA (0.1 ppts of GDP); (iii) postponing transfers to local councils to next year (0.1 ppts of GDP); (iv) freezing domestically financed capital expenditure in Q4 to contain overruns on it at 0.3 percent of GDP, which in addition to the removal of one-off should yield about 2 ppt of GDP reduction relative to 2022; and (v) institute a freeze on new hiring and new wage increases during 2023. In 2024, the deficit is expected to narrow to 2.8 percent of GDP. Revenues are expected to rise gradually over the medium-term supported by the authorities' efforts to boost revenue collection within the context of the MTRS. In particular, the authorities plan to implement additional measures in the 2025 Finance Act to raise another 1 percent of GDP in tax revenues. Meanwhile, limits on discretionary spending, efficiency improvements in capital expenditures, and better commitment management would contain spending. The assumption on domestic arrears clearance remains unchanged from the June 2023 DSA, with the cash paydown of arrears limited to a low budget ceiling and accompanied by NPV reductions in the face value of debt.[4] The deficit is expected to be financed with external concessional financing (both multilateral and bilateral) and domestic market financing.

- **External.** Sierra Leone continues to face strong negative terms-of-trade shocks due to higher fuel and food prices and lower iron-ore prices amid the war in Ukraine and the slowdown in

[3] Preliminary outturn indicates that expenditures have trended slightly above target, at 78.4 percent of year's target during Q1-Q3 (or 19 percent of GDP). However, revenue collection has fallen modestly short at 9.0 percent of GDP during Q1-Q3 or 67 percent of the year's target, largely due to technical delays in implementing the tax policy measures introduced in the 2023 Finance Act.

[4] The stock of domestic legacy arrears (these all are for suppliers of goods and services and are not caused by the government's liquidity problems) amounted to Le 2,581.3 billion (or 4.8 percent of GDP) by end 2022. The authorities have an arrears clearance strategy in place with annual budgeted amounts to be amortized. https://mofsl.gov.sl/wp-content/uploads/2020/09/Sierra-Leone-Arrears-Clearance-Strategy-and-Principles-2020-2025.pdf

China. The current account deficit is projected to shrink in 2023, as increasing iron-ore exports offset imports. The current account is projected to gradually improve over the medium-term, supported by strengthening mining production.

- **IDA financing.** The assumption regarding IDA financing modalities is unchanged. IDA financing reflects firm grants commitments until 2027 and 100 percent regular IDA credits after 2028.[5]

- **The additional use of the 2021 IMF SDR allocation is expected to help safeguard priority spending.** This DSA assumes approximately Le 1,233 billion (2.3 percent of GDP) and Le 1,866 billion (2.5 percent of GDP), to be used in 2022 and 2023. The amounts will be transferred to the government through a retrocession transaction. The SDR assets will then be sold back to the central bank which will convert them into local currency and credit the government's accounts accordingly. The operation would leave the government with SDR liabilities which are counted as additional long-term external debt in the DSA. The government has committed to transparently account for the use of SDRs to finance priority social spending.

- **Other assumptions.** As before, this DSA reflects actual and projected future disbursements under the ECF, the SDR allocation in August 2021, debt relief under the CCRT, and debt deferment under the DSSI. This DSA also takes into account the projected external financing gap during 2024-28 and assumes that the gap will be covered by concessional financing with an overall grant element of 35.5 percent. More specifically, IMF financing under the ECF has a 5.5-year grace period and 10-year maturity and currently carries a zero interest (the interest rate is subject to Executive Board review every two years). Loans from other multilateral creditors other than the World Bank are assumed to have an overall grant element of 35 percent. This assumption is unchanged from the previous DSA. In terms of domestic financing, as the authorities have successfully continued to issue two- to three-year bonds in recent years, this DSA assumes (i) 15 percent and 20 percent of new domestic debts are financed by 1-3 years bonds in 2023, and 2024, respectively, and (ii) 25 percent of new domestic debts are financed by 1-3 years bonds and 5 percent by 4-7 years bonds between 2025-28. This assumption is unchanged from the previous DSA.[6] Cumulative net financing from commercial banks and non-banks is assumed to stay around 2 percentage points of GDP, as banks' investment model remains heavily reliant on government securities.

6. The realism tools flag optimism relative to the historical trend as the required fiscal consolidation effort is large. The 3-year average primary balance adjustment is over 8 ppts of

[5] This treatment of IDA financing is an exception.

[6] In the medium- to long-term, this assumption allows gradual transition towards the financing strategy (strategy 2) proposed by a Fund/World Bank TA on the MTDS (in Oct 2021) which assumes domestic borrowing includes issuing 2-5-year T-bonds over the medium term (share of T-bonds in domestic debt securities will increase to 30 percent by 2025). The authorities adopted an updated 2023-27 MTDS in October 2023. The revised strategy envisages actions to strengthen demand for longer term bonds, to achieve the 40 percent share of T-bonds objective and help bring down elevated levels of debt service, including by (i) stipulating investment norms for institutional investors; (ii) updating auction procedures; (iii) strengthening communication with banks and non-banks on the authorities' MTDS; (iv) reviewing primary dealership roles among commercial banks; and (v) raising limits on T-bond yield.

GDP—close to the 99th percentile of the historical distribution for Sierra Leone's peers. This large adjustment need reflects large fiscal slippages partly due to exogenous shocks (delayed recovery from covid impacts, Russia's invasion of Ukraine, and the cost-of-living crisis), and policy-related slippages, calling for a more frontloaded and urgent fiscal consolidation to keep debt on a downward path (¶5). The authorities have implemented strong corrective measures in the context of the 6th, 7th, and 8th reviews of the Fund-supported program. The package of measures on revenue mobilization, including enforcement and compliance, and expenditure containment and controls is expected to help address past slippages, ringfence implementation of the current year's budget and underpin a sustainable medium-term fiscal framework. The authorities have also identified strong contingency measures. They are prepared to frontload additional revenue measures based on the MTRS, including harmonizing final with holding CIT taxes at a rate of 20 percent, increasing the stamp duty on immovable property and further streamlining GST exemptions. They are also seeking additional budget support grants and would further reprioritize spending, including by curbing domestically financed capital expenditure, should spending pressures reemerge. The authorities are committed to maintaining cost-reflective energy prices by applying the fuel and electricity price formulas, including in case of additional global energy supply shocks. The fiscal drag on near term growth is projected to be cushioned by Sierra Leone's open economy and relatively insulated mining and agricultural sectors.

7. The deterioration of the macroeconomic environment and continued depreciation of the Leone have accentuated the risks around the baseline. Scope to maneuver depends on making progress on planned fiscal adjustment and seeking additional budget support grants. With the measures already introduced (¶5) and expected to be introduced as part of the 2025 Finance Act, domestic revenue is projected to increase over the medium term by more than 4 ppts of GDP. Efficiency gains from digitalization, a gradual recovery in demand and tax buoyancy, and the implementation of the automatic pricing formula for fuel products are also expected to support revenue mobilization. The medium-term wage bill management strategy (to be developed with Fund TA), and reforms on subsidies are expected to help contain recurrent expenditures, while domestic capex is expected to continue supporting human capital development objectives. The authorities remain committed to a concessional debt ceiling and a zero non-concessional borrowing policy and are seeking additional budget support grants for priority and social spending, which will help contain debt vulnerabilities. The authorities have committed to strengthening budget execution, including through regular meetings of the extended cash and debt management committee, ensuring that quarterly budget allotments are backed by credible cash flow projections.

COUNTRY CLASSIFICATION AND DETERMINATION OF SCENARIO STRESS TESTS

8. Sierra Leone's debt-carrying capacity is medium. With the Composite Indicator (CI) score of 2.73 based on the latest data including the April 2023 WEO and the World Bank's latest 2022 CPIA, Sierra Leone's debt carrying capacity is evaluated as medium, unchanged from the previous DSA (Text Tables 4 and 5).

Text Table 4. Composite Indicator

Final	Classification based on current vintage	Classification based on the previous vintage	Classification based on the two previous vintages
Medium	Medium 2.73	Medium 2.73	Medium 2.75

Components	Coefficients (A)	10-year average values (B)	CI Score components (A*B) = (C)	Contribution of components
CPIA	0.385	3.181	1.22	45%
Real growth rate (in percent)	2.719	3.581	0.10	4%
Import coverage of reserves (in percent)	4.052	33.496	1.36	50%
Import coverage of reserves^2 (in percent)	-3.990	11.220	-0.45	-16%
Remittances (in percent)	2.022	5.493	0.11	4%
World economic growth (in percent)	13.520	2.856	0.39	14%
CI Score			2.73	100%
CI rating			Medium	

Text Table 5. Applicable Thresholds for Debt Indicators

PV of PPG external debt in percent of GDP	40%
PV of PPG external debt in percent of exports	180%
PPG external debt service in percent of exports	15%
PPG external debt service in percent of revenue	18%
PV of total public debt in percent of GDP	55%

9. Stress tests generally follow standardized settings, with the addition of two tailored stress tests to capture risks related to contingent liabilities and commodity prices shocks. The combined contingent liabilities stress test is described in ¶2 above. Commodity exports make up a significant part of the export base, leaving it exposed to potential price shocks. The magnitude of shocks used for the commodity price shock stress test are based on the commodity prices outlook prepared by the IMF Research Department.

EXTERNAL AND PUBLIC DEBT SUSTAINABILITY

External DSA

10. Sierra Leone is assessed to be at high risk of external debt distress, and PPG external debt is assessed to be sustainable. This is predicated on the strong fiscal adjustment and continued reliance on concessional financing and grants, and moderately high growth rates. Due to the sharp exchange rate depreciation, the PV of external debt-to-GDP and the debt service-to-revenue ratio have risen, underscoring the importance of continuing to limit new external borrowing and restricting external borrowing to highly concessional loans.

- Under the baseline scenario, both the PV of the PPG external debt-to-GDP ratio and the PV of the PPG external debt-to-export ratio decline over the medium term (Figure 1, Table 1).[7] The projected decline in the debt ratios reflects several factors such as the repayment of past ECF disbursements, including those from the Ebola period, and the projected improvement in GDP and exports. While the historical scenario indicates increasing debt ratios, this reflects the COVID-19 and Ebola crises as well as recent large commodity price shocks.

- The PPG external debt service-to-revenue ratio stays above the threshold over the medium term, indicating a tight liquidity situation, before steadily declining in the medium- to long-term, although it remains significantly above its threshold of 18 percent for the next six years. The PPG external debt service-to-exports ratio stays slightly below the threshold of 15 percent over the medium term. Although FX inflows from external borrowing are limited, with minimal borrowing under the IMF program ceiling, the authorities are utilizing measures to increase the FX inflow to maintain reserve adequacy. As such, while the indicator is above the threshold in the near term, staff assesses gross international reserves as adequate to fulfill upcoming debt service obligations.

- Stress tests indicate that the external debt indicators are sensitive to multiple shocks, particularly exports. Additionally, all debt indicators are above the thresholds over the medium term in response to the alternative shock of historical averages. The PV of external debt to GDP remains above the thresholds for the whole projection period in response to a shock to real GDP, and over the medium term in response to commodity, other flows and depreciation shocks. Lastly, all shock scenarios cause external debt service to revenues to breach the threshold.

- Since the PPG external debt service-to-revenue ratio breaches its threshold over the medium term under the baseline, Sierra Leone is assessed to be at high risk of external debt distress. However, all the external debt indicators are on a declining trend over the medium to long term under the policy settings in the ECF-supported program.

Overall Risk of Public Debt Distress

11. Sierra Leone is assessed to be at high overall risk of public debt distress, but public debt is assessed to be sustainable. Again, this is predicated on the strong and frontloaded fiscal adjustment and continued reliance on concessional financing and grants, and moderately high growth rates.

- Under the baseline scenario, the PV of the public debt-to-GDP ratio breaches the threshold of 55 percent of GDP initially but gradually declines and falls below it by 2027 (Figure 2). The indicator shifted up due to the depreciation of the leone and falls below the threshold one year later than before. The public debt service-to-revenue ratio is projected to decline steadily over the medium term but remains high even at the end of the projection horizon, suggesting a tight

[7] The residuals in Table 1 reflect exchange rate changes; differential between average and period end exchange rate estimates; and decrease in capital flows, implying higher other BOP inflows, including use of reserves.

liquidity situation. Recent efforts to lengthen the maturity structure of government borrowing reduces the ratio. This large debt service is expected to be financed with external grants, concessional loans, and primary surpluses. In this context, Sierra Leone will need continued access to concessional financing to ensure that financing terms remains favorable. In the long term, as the economy recovers and revenue mobilization gains materialize, the public debt service-to-revenue ratio is expected to decline.

- Stress tests indicate that a shock to GDP is the most extreme shock. In response to this shock, the PV of debt to GDP ratio breaches the threshold for the whole projection period, as it does to a commodity shock but at a lower ratio. In fact, all shocks simulated lead the PV of debt to GDP ratio to breach the threshold. Considering that both external debt indicators and public debt indicators exceed their thresholds under the baseline, the country is assessed to have high overall risk of public debt distress.

- In this context, (i) sustained and significant fiscal adjustment; and (ii) continued reliance on highly concessional external financing (ideally grants) are particularly important, including from IFIs which account for a large share of Sierra Leone's PPG external debt.

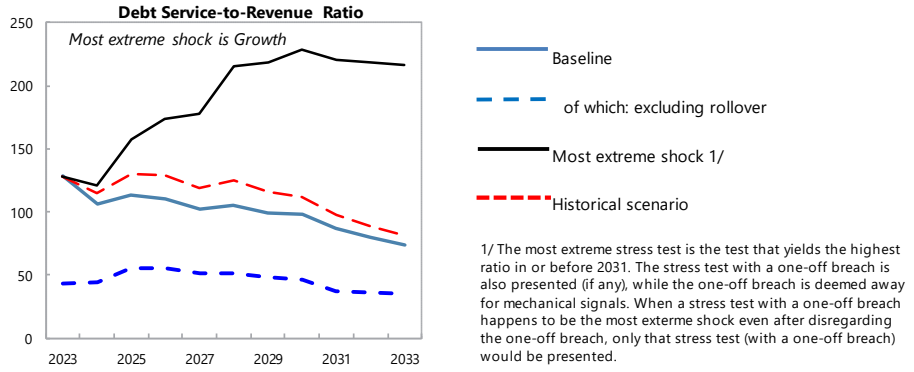
12. The debt service to revenue ratio and gross financing needs are high and their reduction is contingent on the ambitious fiscal adjustment and greater grant and/or concessional borrowing. The need to roll over T-bills issued in the previous year ("T-bills amortization" in Text Table 6) accounts for around 70 percent of gross financing needs (Text Table 6 and Text Figure 3). These T-bills are held largely by domestic public banks and domestic subsidiaries of foreign banks. Rollover risk is high. However, considering the characteristics of Sierra Leone's domestic financial market—where commercial banks' investment models rely heavily on T-Bills, there is limited secondary market trading between commercial banks—staff assesses that these factors mitigate the liquidity risks from this rollover. Staff also assesses that banks have the capacity to absorb the projected volumes assumed under the baseline. Going forward, rollover risks should decline with the reduction in the government's domestic financing needs and the general tightening of macroeconomic policies, which should contain inflation and the depreciation of the leone. In case significant negative shocks materialize, the burden on banks to absorb short-term government debt will rise further, leading to larger rollover risks, especially if banking sector health also deteriorates following the shock. The gross financing needs will increase if the authorities are unable to roll over at reasonable costs. Lengthening of the maturity structure, along with strong fiscal consolidation and efforts to secure more grant financing is imperative.

Text Table 6. Debt-Service to Revenue and Grants Ratio

	Actual	Projections											
	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033	2043
Debt service-to-revenue and grants ratio 1/	**111.0**	**128.3**	**106.6**	**113.3**	**110.5**	**102.0**	**105.1**	**99.4**	**97.7**	**86.8**	**79.3**	**73.3**	**23.0**
of which: T-bills amortization	77.9	84.8	61.9	57.6	55.4	50.5	54.2	51.5	51.7	49.3	43.1	38.2	5.6
of which: others	**33.1**	**43.5**	**44.7**	**55.6**	**55.0**	**51.5**	**50.9**	**47.9**	**46.0**	**37.5**	**36.3**	**35.1**	**17.5**
Gross financing need 2/	**28.0**	**23.9**	**20.3**	**21.6**	**20.5**	**19.5**	**20.2**	**19.2**	**18.8**	**16.7**	**15.2**	**14.1**	**5.3**
of which: T-bills amortization	14.6	14.7	12.5	11.7	10.9	10.1	9.5	9.2	9.1	8.9	7.7	6.8	1.0
of which: others	**13.3**	**9.2**	**7.8**	**9.9**	**9.6**	**9.5**	**10.8**	**10.0**	**9.7**	**7.8**	**7.5**	**7.3**	**4.3**

Sources: Country authorities; and staff estimates and projections.
1/ Debt service is defined as the sum of interest and amortization of medium and long-term, and short-term debt.
2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period and other debt creating/reducing flows.

Text Figure 3. Debt-Service to Revenue and Grants Ratio



1/ The most extreme stress test is the test that yields the highest ratio in or before 2031. The stress test with a one-off breach is also presented (if any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

RISK RATING AND VULNERABILITIES

13. While Sierra Leone is assessed to be at high risk of external and overall public debt distress, its debt is assessed to be sustainable but growing debt sustainability risks call for increased attention. Large fiscal overruns, together with the depreciation of the exchange rate, have increased risks, as a bigger and more frontloaded fiscal adjustment is now necessary to achieve program objectives. The increasing public debt service-to-revenue ratio over the medium term suggests high liquidity-related vulnerabilities. The stress tests also highlight that debt indicators are sensitive to shocks to exports and GDP growth.

14. This DSA underscores the importance of continued fiscal discipline and structural reforms, supported by technical assistance and prudent financing choices. Higher domestic food and fuel prices could exacerbate the already severe burden on the population, deteriorate the budget and external balances, increase costs for businesses, further expand implicit fuel subsidies, provoke social discontent, and put debt sustainability at risk. Given these vulnerabilities to exogenous shocks, reducing debt, and maintaining debt sustainability requires, first and foremost, sustained fiscal adjustment, underpinned by strengthened public financial management. However, to achieve a pace of fiscal adjustment that does not imperil the post-pandemic recovery and allows the country to protect critical social and health spending and to continue addressing its large development needs, it will be vital to rely on highly concessional financing and ideally grants. As debt sustainability risks increase, securing more grant financing is a first order issue. In line with the limit imposed under the ECF-supported program and WB SDFP and performance and policy actions (PPAs), Sierra Leone has a zero ceiling on non-concessional external debt.

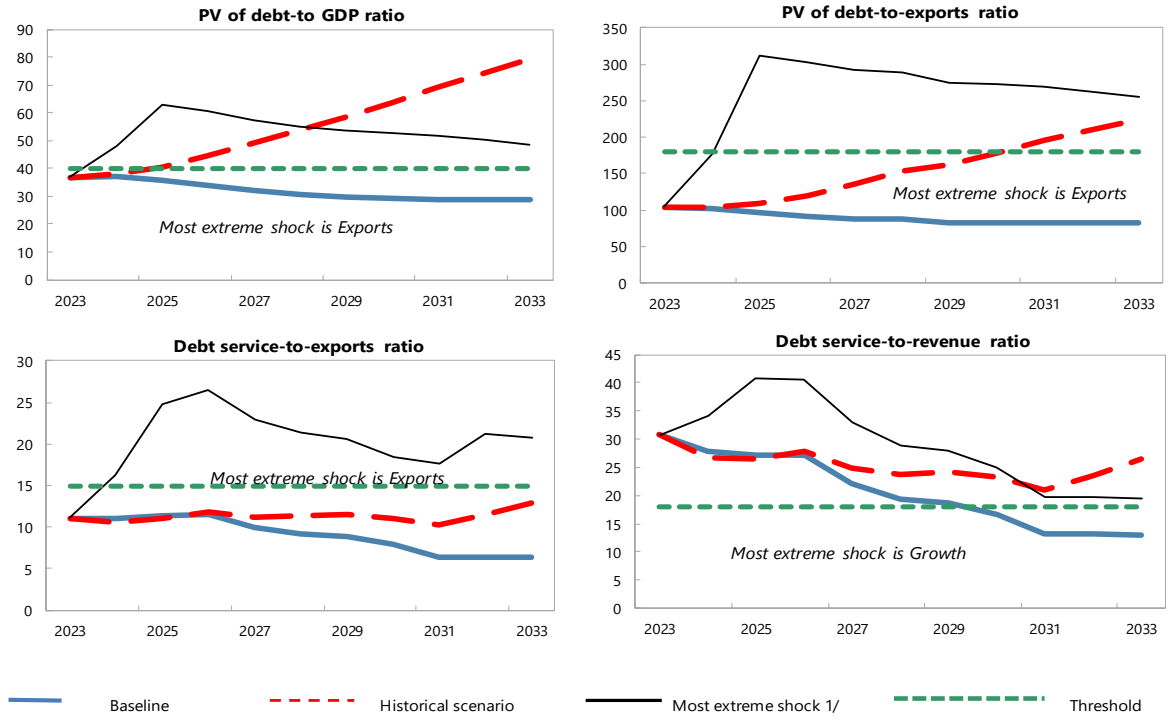
15. Steadfast implementation of the fiscal consolidation, paired with the development of a deeper domestic debt market, will be critical in mitigating rollover risks and maintaining debt sustainability. While staff assesses that risks remain manageable at the current juncture given the characteristics of Sierra Leone's financial market, they could become imminent, and debt burden indicators could shift up considerably, should the authorities be unable to issue medium maturity domestic debt at a reasonable cost, or should the fiscal adjustment fall short of the programmed path, thus further increasing domestic borrowing needs in a context in which commercial banks may

be unable to significantly step-up purchases of government paper. Risks would also increase should the BSL be called upon to finance any additional residual borrowing needs, which would risk reigniting the depreciation and deposit dollarization, thus further constraining banks' domestic currency liquidity. These risks also warrant additional measures including, but not limited to, developing a deeper domestic debt market, together with securing additional grant support. The risks are mitigated by the authorities' commitment to limit future domestic borrowing, as well as continued technical assistance on debt management and the development of the domestic market, including by drawing on recent IMF technical assistance on debt recording, joint World Bank/IMF technical assistance on a medium-term debt strategy, and FY2024 Performance and Policy Actions under Sustainable Development Finance Policy supported by World Bank. Plans for the issuance of medium- to long-term paper would also be essential in extending the yield curve and reducing rollover risks on bonds held by domestic banks.

AUTHORITIES' VIEWS

16. The authorities agreed with staff's views on the DSA. They concurred with the urgency of ensuring a sustained and credible fiscal adjustment in 2023 and onward. They emphasized that they have taken important steps in this direction, included through recently submitted 2024 budget. They committed to continuing to prioritize grant resources, seeking only highly concessional financing, and ratifying only loans within the agreed annual ceiling even after the current IMF ECF program ends to safeguard debt sustainability. The authorities recognized the high T-Bill rollover risks and are committed to addressing these, including by issuing medium-to-long-term bonds and supporting the development of the domestic market, through swift implementation of the recently updated medium-term debt strategy. The authorities reiterated their commitment to clear domestic arrears and are working to address the challenges in obtaining NPV discounts through an updated arrears clearance strategy.

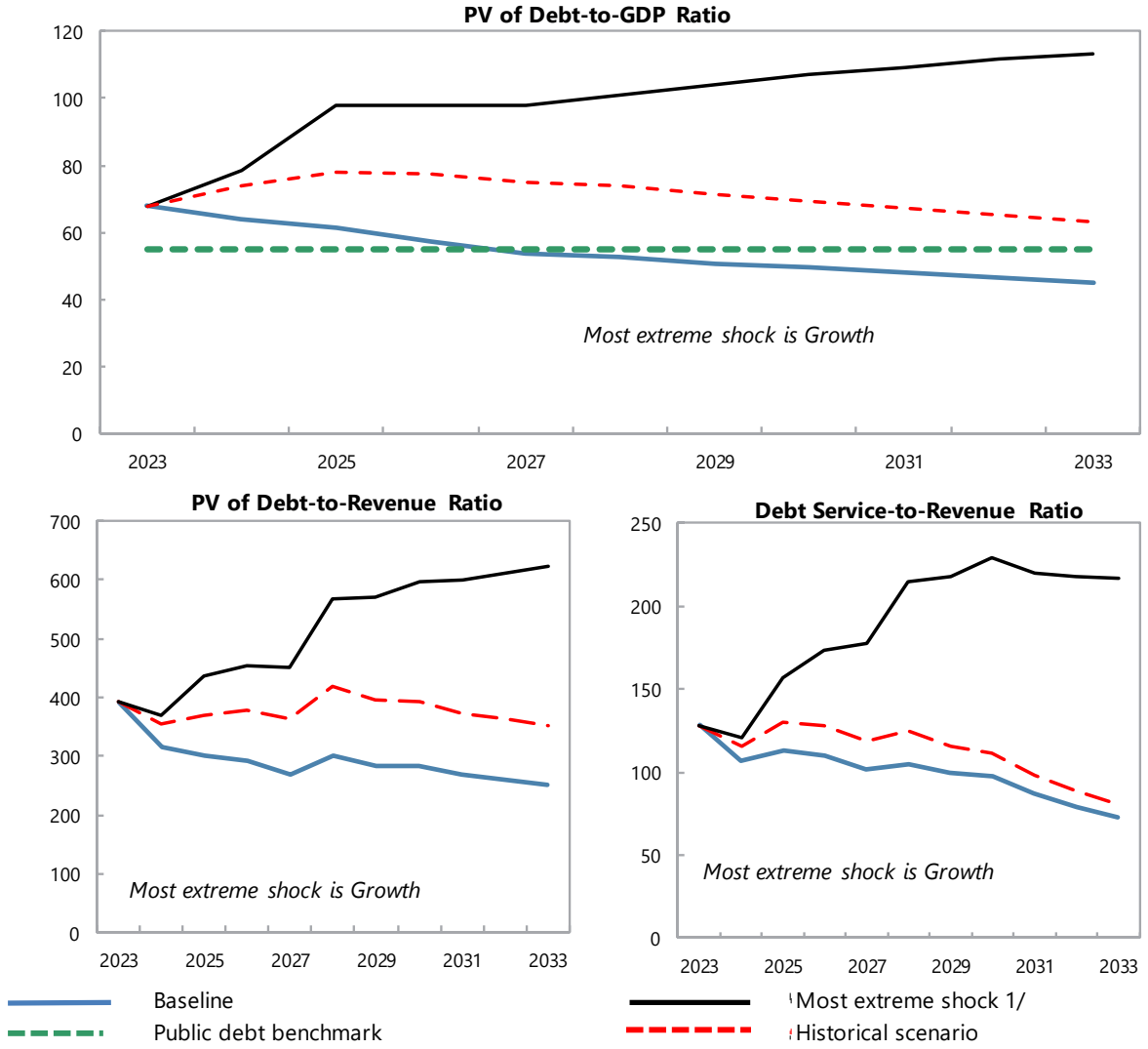
Figure 1. Sierra Leone: Indicators of Public and Publicly Guaranteed External Debt Under Alternative Scenarios, 2023–33



Sources: Sierra Leonean authorities; and IMF staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2032. The stress test with a one-off breach is also presented (in any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most extreme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented. The magnitude of shocks used for the commodity price shock stress test are based on the commodity prices outlook prepared by the IMF research department.

Figure 2. Sierra Leone: Indicators of Public Debt Under Alternative Scenarios, 2023–33



* Note: The public DSA allows for domestic financing to cover the additional financing needs generated by the shocks under the stress tests in the public DSA. Default terms of marginal debt are based on baseline 10-year projections.

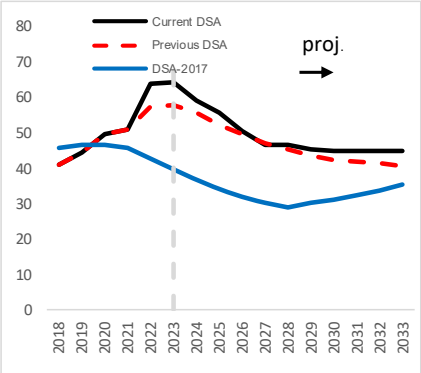
Sources: Sierra Leonean authorities; and IMF staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2032. The stress test with a one-off breach is also presented (in any), while the one-breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most extreme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

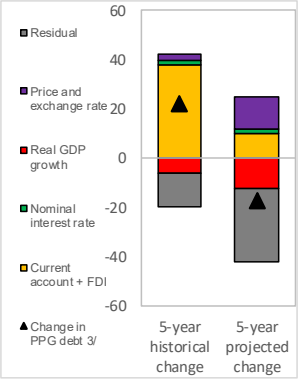
Figure 3. Sierra Leone: Drivers of Debt Dynamics, Baseline Scenario, 2018–33

External debt

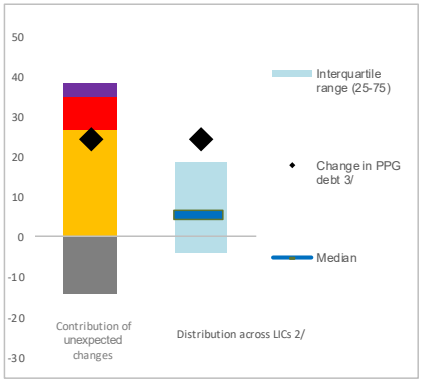
Gross Nominal PPG External Debt
(In percent of GDP; DSA vintages)



Debt-creating flows
(Percent of GDP)

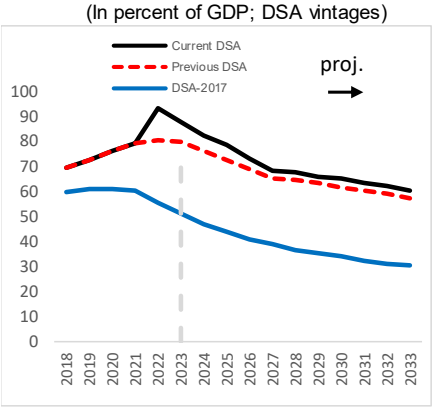


Unexpected Changes in Debt 1/
(Past 5 years, percent of GDP)

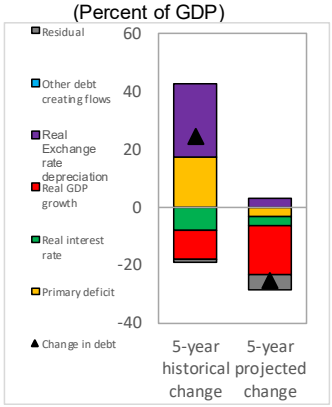


Public debt

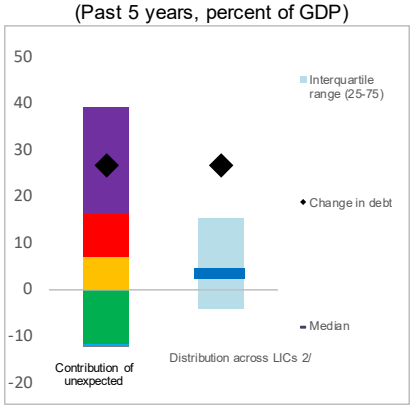
Gross Nominal Public Debt
(In percent of GDP; DSA vintages)



Debt-creating flows
(Percent of GDP)



Unexpected Changes in Debt 1/
(Past 5 years, percent of GDP)



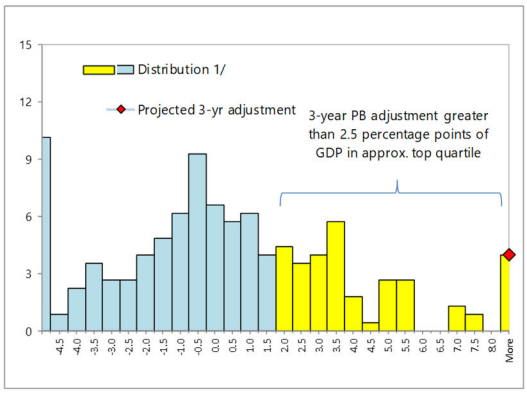
1/ Difference between anticipated and actual contributions on debt ratios.

2/ Distribution across LICs for which LIC DSAs were produced.

3/ Given the relatively low private external debt for average low -income countries, a ppt change in PPG external debt should be largely explained by the drivers of the external debt dynamics equation.

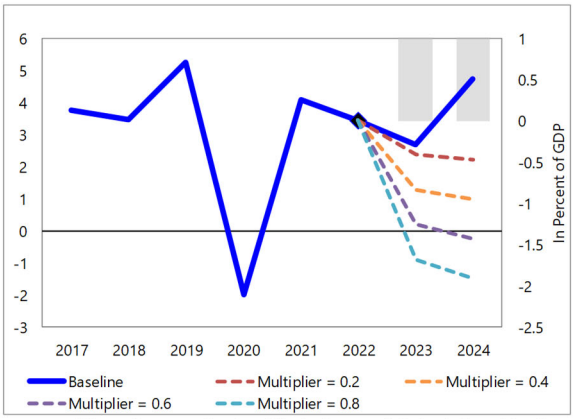
Figure 4. Sierra Leone: Realism Tools

3-Year Adjustment in Primary Balance
(In Percent of GDP)



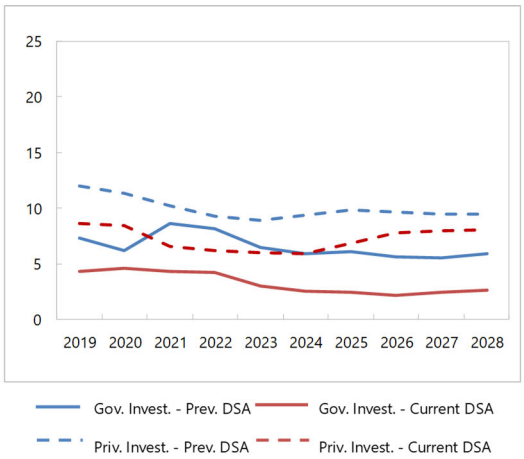
1/ Data cover Fund-supported programs for LICs (excluding emergency financing) approved since 1990. The size of 3-year adjustment from program inception is found on the horizontal axis; the percent of sample is found on the vertical axis.

Fiscal Adjustment and Possible Growth Paths 1/
(Percent)



1/ Bars refer to annual projected fiscal adjustment (right-hand side scale) and lines show possible real GDP growth paths under different fiscal multipliers (left-hand side scale).

Public and Private Investment Rates
(Percent of GDP)



Contribution to Real GDP growth
(Percent, 5-year average)

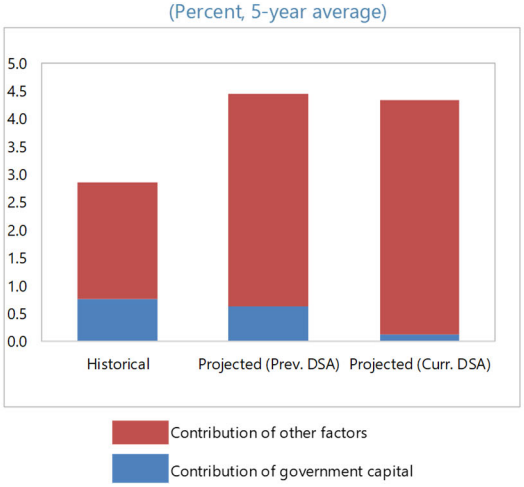
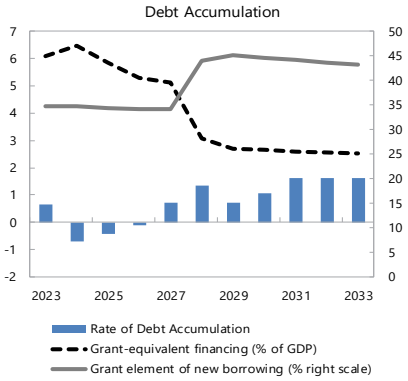


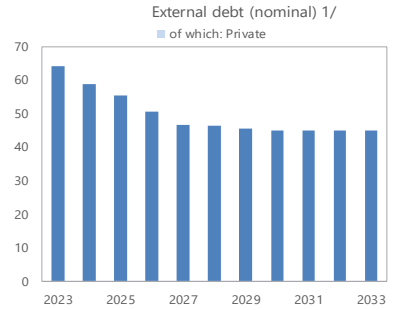
Table 1. Sierra Leone: External Debt Sustainability Framework, Baseline Scenario, 2022–43

(Percent of GDP, unless otherwise indicated)

	Actual	Projections								Average 8/	
	2022	2023	2024	2025	2026	2027	2028	2033	2043	Historical	Projections
External debt (nominal) 1/	**63.8**	**64.1**	**58.9**	**55.4**	**50.6**	**46.7**	**46.5**	**44.9**	**45.2**	**41.0**	**49.7**
of which: public and publicly guaranteed (PPG)	**63.8**	**64.1**	**58.9**	**55.4**	**50.6**	**46.7**	**46.5**	**44.9**	**45.2**	**41.0**	**49.7**
Change in external debt	13.0	0.3	-5.3	-3.4	-4.9	-3.9	-0.2	0.0	0.2		
Identified net debt-creating flows	**8.0**	**3.2**	**0.3**	**-1.1**	**-1.8**	**-1.2**	**-0.7**	**-0.4**	**13.9**	**7.6**	**-0.4**
Non-interest current account deficit	**8.5**	**5.6**	**3.7**	**2.3**	**2.3**	**2.6**	**4.5**	**4.9**	**6.2**	**13.6**	**3.9**
Deficit in balance of goods and services	19.6	16.9	15.7	14.4	14.1	13.9	14.0	13.6	14.1	**21.4**	**14.2**
Exports	30.4	35.3	36.7	37.4	37.0	36.1	35.2	35.3	34.4		
Imports	50.1	52.2	52.4	51.7	51.1	50.0	49.2	48.8	48.5		
Net current transfers (negative = inflow)	-12.3	-12.6	-13.2	-13.2	-13.0	-12.5	-10.6	-9.9	-8.7	**-10.7**	**-11.5**
of which: official	-3.7	-3.0	-3.1	-3.0	-2.9	-2.7	-1.1	-1.0	-0.8		
Other current account flows (negative = net inflow)	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2	0.8	**2.9**	**1.2**
Net FDI (negative = inflow)	**-1.5**	**-0.8**	**-0.7**	**-0.9**	**-2.2**	**-2.2**	**-3.7**	**-3.8**	**9.2**	**-6.1**	**-2.6**
Endogenous debt dynamics 2/	**1.0**	**-1.6**	**-2.7**	**-2.5**	**-1.9**	**-1.6**	**-1.6**	**-1.5**	**-1.5**		
Contribution from nominal interest rate	0.4	0.3	0.4	0.5	0.5	0.4	0.4	0.4	0.4		
Contribution from real GDP growth	-1.8	-1.9	-3.1	-3.0	-2.4	-2.1	-2.0	-1.9	-1.9		
Contribution from price and exchange rate changes	2.4	...	...	...	...	...	...	...	...		
Residual 3/	**5.0**	**-2.9**	**-5.6**	**-2.3**	**-3.0**	**-2.7**	**0.5**	**0.4**	**-13.7**	**-3.8**	**-1.3**
of which: exceptional financing	0.1	0.1	0.1	0.1	0.1	0.0	0.0	0.0	0.0		
Sustainability indicators											
PV of PPG external debt-to-GDP ratio	**32.4**	**36.8**	**37.3**	**35.9**	**34.0**	**31.9**	**30.7**	**28.7**	**29.5**		
PV of PPG external debt-to-exports ratio	**106.7**	**104.4**	**101.6**	**96.1**	**92.0**	**88.3**	**87.4**	**81.4**	**85.9**		
PPG debt service-to-exports ratio	**11.1**	**11.1**	**11.1**	**11.3**	**11.6**	**9.9**	**9.2**	**6.4**	**6.3**		
PPG debt service-to-revenue ratio	**26.7**	**30.8**	**28.0**	**27.2**	**27.1**	**22.0**	**19.3**	**13.0**	**12.6**		
Gross external financing need (Million of U.S. dollars)	424.4	319.7	253.5	207.4	168.1	167.5	185.3	201.7	1890.7		
Key macroeconomic assumptions											
Real GDP growth (in percent)	3.5	2.7	4.7	5.2	4.5	4.5	4.6	4.6	4.5	**2.9**	**4.5**
GDP deflator in US dollar terms (change in percent)	-4.6	-12.5	-7.5	-2.4	0.7	4.3	3.3	1.3	1.3	**-1.4**	**-0.7**
Effective interest rate (percent) 4/	0.7	0.4	0.6	0.8	0.9	0.9	1.0	1.0	1.0	**0.7**	**0.9**
Growth of exports of G&S (US dollar terms, in percent)	11.8	4.2	0.8	4.6	4.2	6.3	5.4	5.7	6.2	**4.6**	**5.1**
Growth of imports of G&S (US dollar terms, in percent)	9.7	-6.3	-2.8	1.4	4.0	6.6	6.3	5.8	5.8	**0.6**	**3.5**
Grant element of new public sector borrowing (in percent)	...	34.7	34.7	34.4	34.2	34.2	44.0	43.1	39.0	**...**	**39.7**
Government revenues (excluding grants, in percent of GDP)	12.6	12.7	14.5	15.5	15.8	16.2	16.8	17.2	17.2	**12.7**	**16.1**
Aid flows (in Million of US dollars) 5/	251.1	169.3	243.7	215.7	193.9	196.4	197.9	166.7	193.4		
Grant-equivalent financing (in percent of GDP) 6/	...	6.1	6.5	5.8	5.3	5.1	3.1	2.5	2.3	**...**	**4.1**
Grant-equivalent financing (in percent of external financing) 6/	...	68.4	81.8	74.5	68.1	65.7	50.5	50.9	46.7	**...**	**60.7**
Nominal GDP (Million of US dollars)	4,095	3,680	3,565	3,662	3,854	4,199	4,539	6,065	10,769		
Nominal dollar GDP growth	-1.3	-10.1	-3.1	2.7	5.2	8.9	8.1	6.0	5.9	**1.3**	**3.8**
Memorandum items:											
PV of external debt 7/	32.4	36.8	37.3	35.9	34.0	31.9	30.7	28.7	29.5		
In percent of exports	106.7	104.4	101.6	96.1	92.0	88.3	87.4	81.4	85.9		
Total external debt service-to-exports ratio	11.1	11.1	11.1	11.3	11.6	9.9	9.2	6.4	6.3		
PV of PPG external debt (in Million of US dollars)	1328.6	1355.0	1329.5	1314.7	1311.1	1338.5	1395.1	1742.6	3182.1		
(PVt-PVt-1)/GDPt-1 (in percent)		0.6	-0.7	-0.4	-0.1	0.7	1.3	1.6	2.0		
Non-interest current account deficit that stabilizes debt ratio	-4.5	5.3	9.0	5.8	7.2	6.5	4.7	4.9	6.0		

Definition of external/domestic debt	Residency-based
Is there a material difference between the two criteria?	No





Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g) + Ɛα (1+r)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, ρ = growth rate of GDP deflator in U.S. dollar terms, Ɛ=nominal appreciation of the local currency, and α= share of local currency-denominated external debt in total external debt.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Current-year interest payments divided by previous period debt stock.

5/ Defined as grants, concessional loans, and debt relief.

6/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

7/ Assumes that PV of private sector debt is equivalent to its face value.

8/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

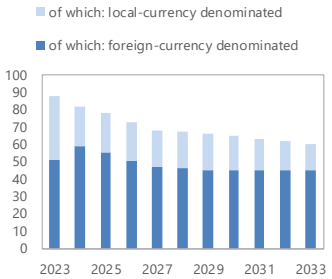
Table 2. Sierra Leone: Public Sector Debt Sustainability Framework, Baseline Scenario, 2022–43

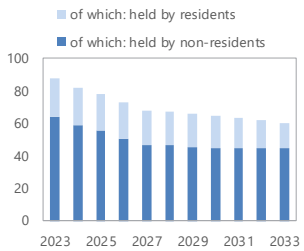
(Percent of GDP, unless otherwise indicated)

	Actual	Projections								Average 6/	
	2022	2023	2024	2025	2026	2027	2028	2033	2043	Historical	Projections
Public sector debt 1/	**93.3**	**87.9**	**82.0**	**78.4**	**73.1**	**68.1**	**67.4**	**60.4**	**47.8**	**63.4**	**70.3**
of which: external debt	**63.8**	**64.1**	**58.9**	**55.4**	**50.6**	**46.7**	**46.5**	**44.9**	**45.2**	**41.0**	**49.7**
Change in public sector debt	14.0	-5.4	-5.9	-3.6	-5.4	-5.0	-0.7	-1.5	-0.9		
Identified debt-creating flows	**18.7**	**-7.1**	**-5.6**	**-4.4**	**-3.4**	**-2.9**	**-0.6**	**-1.8**	**-1.2**	**4.7**	**-2.9**
Primary deficit	**7.1**	**1.7**	**-1.2**	**-1.4**	**-1.3**	**-0.8**	**1.9**	**1.0**	**1.2**	**3.8**	**0.5**
Revenue and grants	18.8	17.3	20.2	20.3	19.8	19.9	17.5	17.9	17.8	**16.7**	**18.6**
of which: grants	6.1	4.6	5.7	4.8	4.0	3.7	0.7	0.7	0.6		
Primary (noninterest) expenditure	25.9	19.0	19.0	18.9	18.5	19.1	19.4	18.9	19.0	**20.4**	**19.0**
Automatic debt dynamics	**11.6**	**-8.8**	**-4.3**	**-3.1**	**-2.1**	**-2.1**	**-2.4**	**-2.7**	**-2.4**		
Contribution from interest rate/growth differential	-8.2	-8.8	-4.3	-3.1	-2.1	-2.1	-2.4	-2.7	-2.4		
of which: contribution from average real interest rate	-5.6	-6.4	-0.4	1.0	1.3	1.0	0.6	0.0	-0.3		
of which: contribution from real GDP growth	-2.6	-2.4	-4.0	-4.0	-3.4	-3.1	-3.0	-2.7	-2.1		
Contribution from real exchange rate depreciation	19.8	...	...	...	...	...	...	...	...		
Other identified debt-creating flows	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**
Privatization receipts (negative)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Recognition of contingent liabilities (e.g., bank recapitalization)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Debt relief (HIPC and other)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Other debt creating or reducing flow (please specify)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Residual	**-4.7**	**1.7**	**-0.4**	**0.9**	**-2.0**	**-2.0**	**-0.1**	**0.2**	**0.4**	**1.0**	**-0.1**
Sustainability indicators											
PV of public debt-to-GDP ratio 2/	**73.0**	**67.8**	**63.8**	**61.4**	**57.5**	**53.7**	**52.5**	**45.0**	**32.9**		
PV of public debt-to-revenue and grants ratio	**388.9**	**392.0**	**315.5**	**302.3**	**291.0**	**269.4**	**300.1**	**251.2**	**184.4**		
Debt service-to-revenue and grants ratio 3/	**111.0**	**128.3**	**106.6**	**113.3**	**110.5**	**102.0**	**105.1**	**73.3**	**23.0**		
Gross financing need 4/	28.0	23.9	20.3	21.6	20.5	19.5	20.2	14.1	5.3		
Key macroeconomic and fiscal assumptions											
Real GDP growth (in percent)	3.5	2.7	4.7	5.2	4.5	4.5	4.6	4.6	4.5	**2.9**	**4.5**
Average nominal interest rate on external debt (in percent)	0.8	0.6	0.7	0.9	0.9	1.0	1.0	1.0	1.0	**0.7**	**0.9**
Average real interest rate on domestic debt (in percent)	-9.9	-14.5	2.1	8.6	8.7	7.0	4.8	2.4	2.3	**-0.4**	**2.8**
Real exchange rate depreciation (in percent, + indicates depreciation)	42.9	...	...	...	...	...	...	...	...	**8.2**	**...**
Inflation rate (GDP deflator, in percent)	25.3	37.1	21.7	14.7	10.9	8.7	7.9	6.4	6.3	**10.8**	**12.1**
Growth of real primary spending (deflated by GDP deflator, in percent)	5.5	-24.7	4.8	4.7	2.1	8.1	5.9	4.3	4.6	**6.7**	**1.9**
Primary deficit that stabilizes the debt-to-GDP ratio 5/	-6.9	7.1	4.7	2.2	4.1	4.2	2.6	2.5	2.1	**-2.2**	**3.4**
PV of contingent liabilities (not included in public sector debt)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		

Definition of external/domestic debt	**Residency-based**
Is there a material difference between the two criteria?	**No**

Public sector debt 1/





Sources: Country authorities; and staff estimates and projections.

1/ Coverage of debt: The central government plus social security, central bank, government-guaranteed debt. Definition of external debt is Residency-based.

2/ The underlying PV of external debt-to-GDP ratio under the public DSA differs from the external DSA with the size of differences depending on exchange rates projections.

3/ Debt service is defined as the sum of interest and amortization of medium and long-term, and short-term debt.

4/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period and other debt creating/reducing flows.

5/ Defined as a primary deficit minus a change in the public debt-to-GDP ratio ((-): a primary surplus), which would stabilizes the debt ratio only in the year in question.

6/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

Note: Grants include grants for debt service relief under the Catastrophe Containment and Relief Trust (CCRT).

Table 3. Sierra Leone: Sensitivity Analysis, External Debt, 2023-33

	Projections 1/										
	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033
PV of debt-to GDP ratio											
Baseline	36.8	37.3	35.9	34.0	31.9	30.7	29.7	29.0	28.9	28.8	28.7
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2023-2033 2/	36.8	38.0	**40.6**	**44.4**	**49.0**	**53.9**	**58.5**	**63.6**	**69.0**	**74.2**	**79.2**
B. Bound Tests											
B1. Real GDP growth	36.8	**45.6**	**53.8**	**51.0**	**47.7**	**46.0**	**44.5**	**43.5**	**43.3**	**43.2**	**43.0**
B2. Primary balance	36.8	38.2	38.4	37.5	36.1	35.5	34.9	34.6	34.7	34.6	34.5
B3. Exports	36.8	**48.2**	**63.0**	**60.6**	**57.2**	**55.2**	**53.7**	**52.6**	**51.7**	**50.2**	**48.7**
B4. Other flows 3/	36.8	**43.9**	**49.2**	**47.2**	**44.5**	**42.9**	**41.7**	**40.8**	**40.2**	39.2	38.2
B5. One-time 30 percent nominal depreciation	36.8	**48.2**	**43.8**	**41.4**	38.7	37.3	36.0	35.2	35.1	35.1	35.2
B6. Combination of B1-B5	36.8	**48.9**	**57.8**	**55.3**	**52.1**	**50.3**	**48.8**	**47.8**	**46.9**	**45.9**	**44.9**
C. Tailored Tests											
C1. Combined contingent liabilities	36.8	38.8	38.3	37.2	35.5	34.7	33.9	33.5	33.5	33.5	33.5
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	36.8	**42.8**	**46.7**	**45.0**	**42.5**	**40.7**	39.1	37.7	36.7	35.4	34.1
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	40	40	40	40	40	40	40	40	40	40	40
PV of debt-to-exports ratio											
Baseline	104.4	101.6	96.1	92.0	88.3	87.4	82.2	81.6	81.8	81.5	81.4
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2023-2033 2/	104.4	103.5	108.5	120.0	135.7	153.2	161.8	178.6	**195.2**	**209.8**	**224.5**
B. Bound Tests											
B1. Real GDP growth	104.4	101.6	96.1	92.0	88.3	87.4	82.2	81.6	81.8	81.5	81.4
B2. Primary balance	104.4	104.3	102.8	101.5	100.1	100.9	96.7	97.2	98.1	98.0	97.9
B3. Exports	104.4	177.4	**310.9**	**302.1**	**292.3**	**289.2**	**273.9**	**272.5**	**269.8**	**261.5**	**254.5**
B4. Other flows 3/	104.4	119.6	131.8	127.6	123.3	122.0	115.4	114.7	113.7	110.8	108.4
B5. One-time 30 percent nominal depreciation	104.4	101.6	90.6	86.5	82.9	82.0	77.1	76.4	76.8	76.8	77.1
B6. Combination of B1-B5	104.4	146.3	126.5	172.6	166.7	164.9	155.8	154.9	153.2	149.7	146.8
C. Tailored Tests											
C1. Combined contingent liabilities	104.4	105.8	102.6	100.5	98.3	98.5	93.9	94.0	94.8	94.8	95.0
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	104.4	150.7	153.2	143.1	132.9	125.6	112.6	110.3	107.9	103.9	100.6
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	180	180	180	180	180	180	180	180	180	180	180
Debt service-to-exports ratio											
Baseline	11.1	11.1	11.3	11.6	9.9	9.2	8.9	7.9	6.4	6.4	6.4
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2023-2033 2/	11.1	10.6	11.0	11.9	11.2	11.3	11.5	11.0	10.2	11.5	12.9
B. Bound Tests											
B1. Real GDP growth	11.1	11.1	11.3	11.6	9.9	9.2	8.9	7.9	6.4	6.4	6.4
B2. Primary balance	11.1	11.1	11.3	11.7	10.1	9.4	9.1	8.1	6.8	7.0	7.1
B3. Exports	11.1	**16.3**	**24.8**	**26.5**	**22.9**	**21.3**	**20.6**	**18.4**	**17.6**	**21.2**	**20.7**
B4. Other flows 3/	11.1	11.1	11.6	12.2	10.5	9.8	9.4	8.4	8.0	8.9	8.7
B5. One-time 30 percent nominal depreciation	11.1	11.1	11.3	11.5	9.8	9.1	8.8	7.8	6.4	6.0	6.0
B6. Combination of B1-B5	11.1	13.2	**16.4**	**17.0**	14.7	13.7	13.2	11.8	11.4	12.0	11.8
C. Tailored Tests											
C1. Combined contingent liabilities	11.1	11.1	11.4	11.7	10.0	9.4	9.1	8.1	6.6	6.6	6.6
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	11.1	14.5	14.5	14.7	12.3	11.1	10.3	9.1	8.3	9.0	8.8
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	15	15	15	15	15	15	15	15	15	15	15
Debt service-to-revenue ratio											
Baseline	**30.8**	**28.0**	**27.2**	**27.1**	**22.0**	**19.3**	**18.7**	16.7	13.2	13.1	13.0
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2023-2033 2/	**30.8**	**26.7**	**26.6**	**27.9**	**24.8**	**23.6**	**24.2**	**23.2**	**20.9**	**23.5**	**26.4**
B. Bound Tests											
B1. Real GDP growth	**30.8**	**34.2**	**40.7**	**40.6**	**33.0**	**28.9**	**28.0**	**25.0**	**19.7**	**19.7**	**19.5**
B2. Primary balance	**30.8**	**28.0**	**27.3**	**27.4**	**22.4**	**19.7**	**19.1**	17.2	14.0	14.4	14.6
B3. Exports	**30.8**	**30.6**	**32.5**	**33.8**	**27.7**	**24.3**	**23.5**	**21.1**	**19.5**	**23.5**	**23.0**
B4. Other flows 3/	**30.8**	**28.0**	**27.9**	**28.6**	**23.3**	**20.4**	**19.8**	17.7	16.4	**18.3**	17.9
B5. One-time 30 percent nominal depreciation	**30.8**	**36.2**	**35.2**	**34.8**	**28.2**	**24.7**	**24.0**	**21.4**	16.9	16.0	15.9
B6. Combination of B1-B5	**30.8**	**30.4**	**34.3**	**34.6**	**28.3**	**24.8**	**24.0**	**21.5**	**20.3**	**21.4**	**21.0**
C. Tailored Tests											
C1. Combined contingent liabilities	**30.8**	**28.0**	**27.4**	**27.4**	**22.4**	**19.6**	**19.1**	17.1	13.6	13.5	13.4
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	**30.8**	**37.2**	**36.7**	**37.7**	**28.9**	**23.8**	**21.8**	**18.5**	16.4	17.7	17.3
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	18	18	18	18	18	18	18	18	18	18	18

Sources: Country authorities; and staff estimates and projections.
1/ A bold value indicates a breach of the threshold.
2/ Variables include real GDP growth, GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
3/ Includes official and private transfers and FDI.

Table 4. Sierra Leone: Sensitivity Analysis, Public Debt, 2023-33

	Projections 1/										
	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033
	PV of Debt-to-GDP Ratio										
Baseline	**67.8**	**63.8**	**61.4**	**57.5**	53.7	52.5	50.9	49.7	48.1	46.6	45.0
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2023-2033 2/	**68**	**74**	**78**	**78**	**75**	**74**	**71**	**69**	**67**	**65**	**63**
B. Bound Tests											
B1. Real GDP growth	**68**	**79**	**98**	**98**	**98**	**101**	**104**	**107**	**109**	**111**	**113**
B2. Primary balance	**68**	**70**	**74**	**69**	**65**	**63**	**60**	**59**	**57**	55	53
B3. Exports	**68**	**71**	**81**	**76**	**71**	**70**	**68**	**66**	**64**	**61**	**59**
B4. Other flows 3/	**68**	**71**	**76**	**71**	**67**	**65**	**63**	**62**	**60**	**57**	55
B5. One-time 30 percent nominal depreciation	**68**	**66**	**63**	**58**	53	51	48	46	44	42	39
B6. Combination of B1-B5	**68**	**72**	**74**	**64**	**62**	**62**	**61**	**60**	**59**	**58**	**57**
C. Tailored Tests											
C1. Combined contingent liabilities	**68**	**74**	**71**	**66**	**62**	**60**	**58**	**56**	55	53	51
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	**68**	**70**	**75**	**81**	**85**	**91**	**96**	**99**	**102**	**105**	**108**
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Public debt benchmark	55	55	55	55	55	55	55	55	55	55	55
	PV of Debt-to-Revenue Ratio										
Baseline	392.0	315.5	302.3	291.0	269.4	300.1	284.1	282.7	268.2	259.8	251.2
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2023-2033 2/	392	354	369	378	364	418	396	393	373	362	352
B. Bound Tests											
B1. Real GDP growth	392	368	435	454	452	568	570	598	599	612	623
B2. Primary balance	392	347	366	351	324	358	337	334	316	306	296
B3. Exports	392	352	399	386	358	399	379	378	358	342	327
B4. Other flows 3/	392	351	373	360	334	372	353	352	332	319	306
B5. One-time 30 percent nominal depreciation	392	333	313	296	269	291	270	263	245	233	221
B6. Combination of B1-B5	392	352	354	319	303	350	338	341	328	322	315
C. Tailored Tests											
C1. Combined contingent liabilities	392	367	349	335	309	343	324	321	304	295	285
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	392	412	442	493	489	577	560	565	570	587	603
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
	Debt Service-to-Revenue Ratio										
Baseline	128.3	106.6	113.3	110.5	102.0	105.1	99.4	97.7	86.8	79.3	73.3
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2023-2033 2/	128	115	130	129	119	125	116	111	98	88	81
B. Bound Tests											
B1. Real GDP growth	128	121	157	174	178	215	218	229	220	218	216
B2. Primary balance	128	107	135	153	137	137	125	119	104	94	86
B3. Exports	128	107	114	112	103	107	101	99	90	86	80
B4. Other flows 3/	128	107	114	112	103	106	100	99	90	84	78
B5. One-time 30 percent nominal depreciation	128	103	111	107	99	101	96	94	83	76	70
B6. Combination of B1-B5	128	109	122	120	116	127	124	126	116	110	105
C. Tailored Tests											
C1. Combined contingent liabilities	128	107	147	141	126	127	117	112	98	89	81
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	128	132	140	155	173	211	212	216	212	212	213
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.

Sources: Country authorities; and staff estimates and projections.
1/ A bold value indicates a breach of the threshold.
2/ Variables include real GDP growth, GDP deflator and primary deficit in percent of GDP.
3/ Includes official and private transfers and FDI.